

Multiple Choice (10 marks)

1. What was the first buffer overflow attack?
 - (a) Love Bug
 - (b) SQL Slammer
 - (c) Morris Worm
 - (d) Code Red
2. How do you prevent SQL injection?
 - (a) Escape queries
 - (b) Interrupt requests
 - (c) Merge tables
 - (d) All of the above
3. Public key encryption is advantageous over Symmetric key Cryptography because of
 - (a) Speed
 - (b) Space
 - (c) Key exchange
 - (d) Key length
4. Which form of encryption does WPA use?
 - (a) Shared key
 - (b) LEAP
 - (c) TKIP
 - (d) AES
5. Which among the following is the least strong security encryption standard?
 - (a) WEP
 - (b) WPA
 - (c) WPA 2
 - (d) WPA 3

True / False (10 marks)

Answer True or False

6. True or False: Buffer overflow vulnerabilities in C and C++ arise primarily from insufficient database checks during application execution.
7. True or False: DNSlookup is commonly used for SNMP enumeration in network security assessments.
8. True or False: WPA primarily relies on AES encryption for securing wireless communication.
9. True or False: Penetration testing is primarily a procedure for testing individual program components for vulnerabilities.
10. True or False: A secure pseudorandom permutation (PRP) can be constructed from a secure pseudorandom generator (PRG) using the GGM PRF and Luby-Rackoff theorem.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to count the number of rectangles in a circle of radius r .
12. Write a python function to count numeric values in a given string.

End of Paper